

Deploying a Scalable Web Application on AWS using Auto Scaling & Load Balancer (Step-by-Step Guide)

Introduction

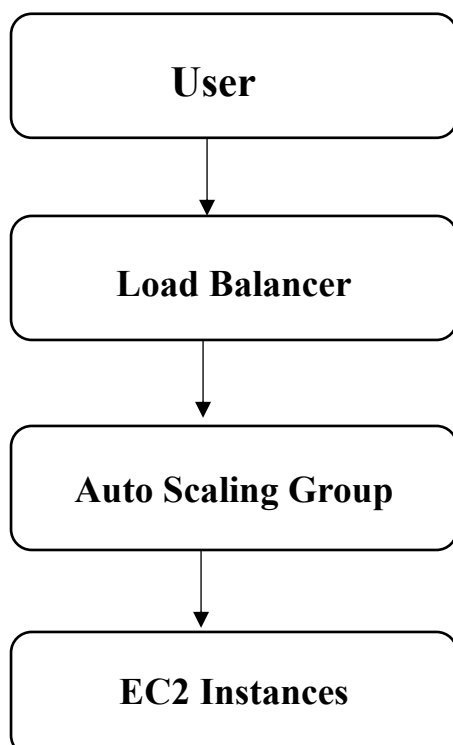
Modern web applications must be scalable, highly available, and fault-tolerant. This guide demonstrates how to design and deploy a simple web application on AWS using an Auto Scaling Group (ASG) behind an Elastic Load Balancer (ELB)

Key Concepts

- **Auto Scaling**
Auto Scaling automatically adjusts the number of EC2 instances in response to demand, ensuring consistent performance at optimal cost.
- **Load Balancer**
A Load Balancer distributes incoming application traffic across multiple targets (such as EC2 instances) in multiple Availability Zones.

In simple terms, Auto Scaling adds more servers when traffic increases and removes them when traffic decreases and Load Balancer acts like a traffic controller that ensures no single server is overloaded.

Request Flow

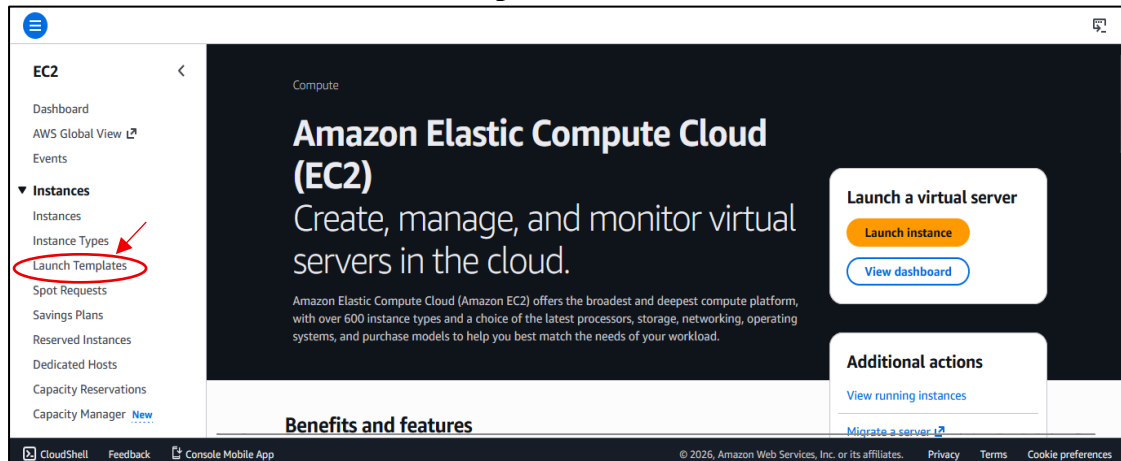


Step-by-Step Implementation

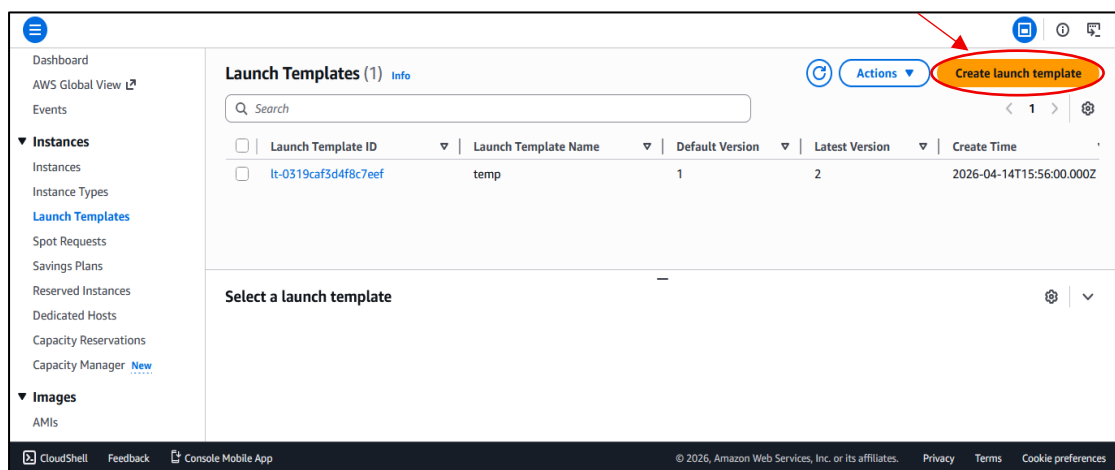
Step 1 : Create a Launch Template

A Launch Template defines how EC2 instances should be created automatically.

1. Go to Amazon EC2 → Launch Templates



2. Click Create Launch Template



3. Configure Template

- **Template name:** (any name)

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

Template

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@'.

- **AMI:** Select your created AMI

▼ **Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

🔍 Search our full catalog including 1000s of application and OS images

Recents Quick Start

Don't include in launch template

Amazon Linux
aws

macOS
Mac

Ubuntu
ubuntu

Windows
Microsoft

Red Hat
Red Hat

SUSE L
SUSE

🔍
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-02eb0c2388ee999f9 (64-bit (x86), uefi-preferred) / ami-00ae76dfb904473c4 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible

- **Instance type:** t3.micro (or available alternative)

▼ **Instance type** [Info](#) | [Get advice](#) Advanced

Instance type

t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0112 USD per Hour
On-Demand SUSE base pricing: 0.0112 USD per Hour
On-Demand Windows base pricing: 0.0204 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour
On-Demand RHEL base pricing: 0.04 USD per Hour

Free tier eligible

☐ All generations
[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

- **Key pair:** Select existing or create a new one

▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

KeyMb

🔄 Create new key pair

🔍

Specify a custom value...

Don't include in launch template

KeyMb
Type: rsa

🔄 Create new subnet

4. Network Settings

Security Group:

- Allow HTTP (Port 80)
- Allow SSH (Port 22)

▼ Network settings Info

Subnet Info

Don't include in launch template

When you specify a subnet, a network interface is automatically added to your template.

Create new subnet

Availability Zone Info

Don't include in launch template

Enable additional zones

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Select existing security group

Create security group

Security groups Info

Select security groups

Compare security group rules

WebServerGroup sg-0857c85b46dd88671

VPC: vpc-0b9967b1715ffa936

5. Add User Data Script in Advanced Details

User data - optional Info

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
sudo su
dnf install httpd -y
systemctl start httpd
echo "Hello from user">/var/www/html/index.html
```

6. Click Create Launch Template

EC2 > Launch templates > Create launch template

Choose file

```
#!/bin/bash
sudo su
dnf install httpd -y
systemctl start httpd
echo "Hello from user">/var/www/html/index.html
```

☐ User data has already been base64 encoded

▼ Summary

Software Image (AMI)

Amazon Linux 2023 AMI 2023.11....read more

ami-02eb0c2388ee999f9

Virtual server type (instance type)

t3.micro

Firewall (security group)

WebServerGroup

Storage (volumes)

1 volume(s) - 8 GiB

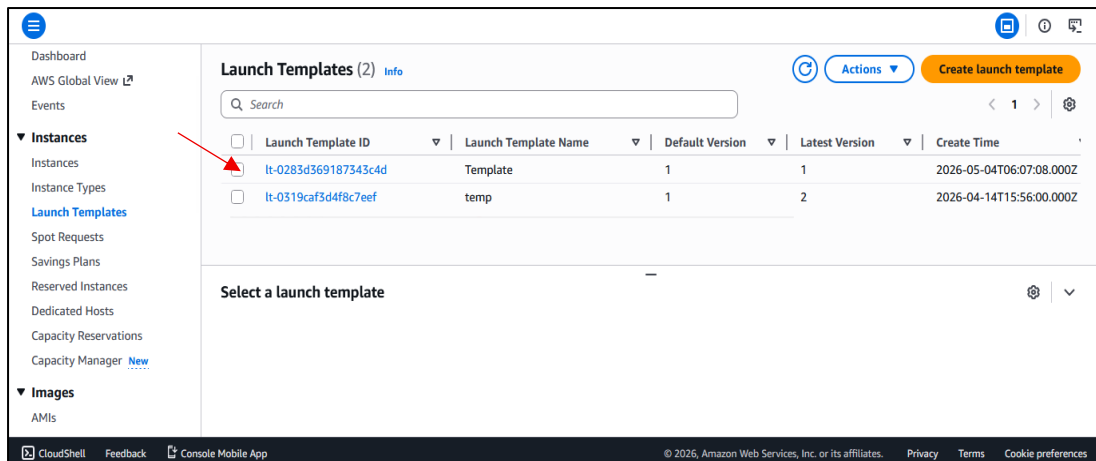
Cancel

Create launch template

CloudShell Feedback Console Mobile App

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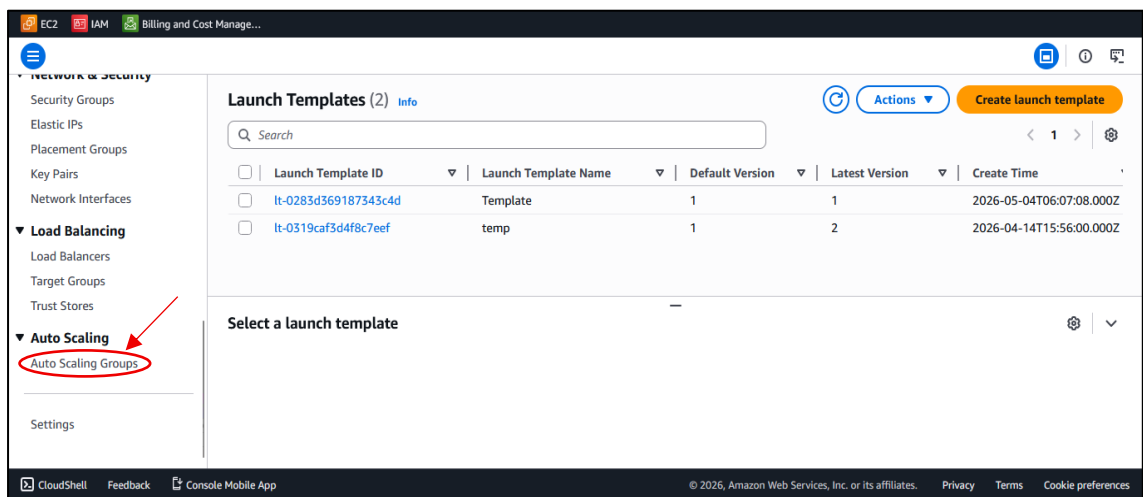
Template has been launched successfully.



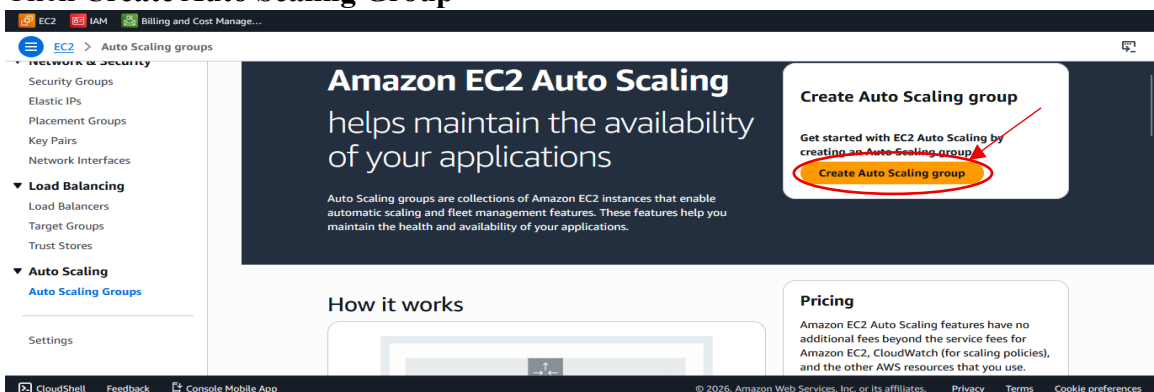
Step 2 : Create an Auto Scaling Group

An Auto Scaling Group automatically launch instances, maintains desired capacity and distributes traffic via Load Balancer.

1. Go to Auto Scaling Groups



2. Click Create Auto Scaling Group



3. Select your Launch Template

Name
Auto Scaling group name
Enter a name to identify the group.

Must be unique to this account in the current Region and no more than 255 characters.

Launch template [Info](#)

For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

[Create a launch template](#)

4. Configure Network Settings

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.

172.31.0.0/16 Default

[Create a VPC](#)

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

☒ **aps1-az1 (ap-south-1a) | subnet-01fb7db6f345e9ca3**
172.31.32.0/20 Default

☒ **aps1-az2 (ap-south-1c) | subnet-0807633b9dd514b5f**
172.31.16.0/20 Default

☒ **aps1-az3 (ap-south-1b) | subnet-07bb40bbd9bb6cb94**
172.31.0.0/20 Default

[Create a subnet](#)

Availability Zone distribution - new
Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

☐ **Balanced only**
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

5. Configure Load Balancer

Load balancing [Info](#)
Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

☐ **No load balancer**
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☐ **Attach to an existing load balancer**
Choose from your existing load balancers.

☒ **Attach to a new load balancer**
Quickly create a basic load balancer to attach to your Auto Scaling group.

Load balancer type
Choose from the load balancer types offered below. Type selection cannot be changed after the load balancer is created. If you need a different type of load balancer than those offered here, [visit the Load Balancing console](#). [L](#)

☒ Application Load Balancer
HTTP, HTTPS

☐ Network Load Balancer
TCP, UDP, TLS

Load balancer name
Name cannot be changed after the load balancer is created.

AsGroupLB

Must be unique within the Region. Use 1–32 alphanumeric characters and hyphens. Can't start with "internal-" or begin or end with a hyphen.

Load balancer scheme
Scheme cannot be changed after the load balancer is created.

☒ Internal

☐ Internet-facing

Network mapping
Your new load balancer will be created using the same VPC and Availability Zone selections as your Auto Scaling group. You can select different subnets and add subnets from additional Availability Zones.

6. Configure Target Group

Listeners and routing
If you require secure listeners, or multiple listeners, you can configure them from the [Load Balancing console](#) [L](#) after your load balancer is created.

Protocol HTTP **Port** 80

Default routing (forward to)
Select new or existing target group

Tags - optional
Consider adding tags to your load balancer. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Add tag](#)

50 remaining

Listeners and routing
If you require secure listeners, or multiple listeners, you can configure them from the [Load Balancing console](#) [L](#) after your load balancer is created.

Protocol HTTP **Port** 80

Default routing (forward to)
Create a target group

New target group name
An instance target group with default settings will be created.

AsGroupTG

Must be unique within the Region. Use 1–32 alphanumeric characters and hyphens. Can't begin or end with a hyphen.

Tags - optional
Consider adding tags to your load balancer. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Add tag](#)

50 remaining

7. Configure VPC Lattice

VPC Lattice integration options [Info](#)

To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.

Select VPC Lattice service to attach

☒ No VPC Lattice service
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.

☐ Attach to VPC Lattice service
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.

[Create new VPC Lattice service](#) [L](#)

8. Configure **Group Size** and **Scaling**

- Set **Desired capacity**

Configure group size and scaling - optional [Info](#)

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [Info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances) ▼

Desired capacity

Specify your group size.

3

- Set **Min Desired Capacity** and **Max Desired Capacity**

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

Max desired capacity

3

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a **target tracking policy** | [Info](#)
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☒ **No scaling policies**

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☐ **Target tracking scaling policy**

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

9. Create Auto Scaling Group

Auto Scaling groups (1) [Info](#)

Last updated less than a minute ago

[Launch configurations](#) [Launch templates](#) [Actions](#) [Create Auto Scaling group](#)

< 1 >

☐	Name	Status - new	Launch template/configuration	Instances	Instance health - ne
☒	AsGroup mb	At desired capacity	Template Version Default	3	3/3 Healthy

10. Verifying Instances, Load Balancer and Target Group

Instances (3) [Info](#) [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets
Choose filter set

Find Instance by attribute or tag (case-sensitive)

< 1 >

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☐		i-0474ee2629b34b255	Running	t3.micro	Initializing	View alarms +	ap-south-1b
☐		i-0febf5886a2a6bee5	Running	t3.micro	Initializing	View alarms +	ap-south-1c
☐		i-0182f4d9f32ae1595	Running	t3.micro	Initializing	View alarms +	ap-south-1a

Load balancers (1) [What's new?](#) [Refresh](#) [Actions](#) [Create load balancer](#)

Filter load balancers

< 1 >

☐	Name	State	Type	Scheme	IP address type	VPC ID
☐	AsGroupLB	Active	application	Internal	IPv4	vpc-0b9967b1715ffa936

Target groups (1) [Info](#) [What's new?](#) [Refresh](#) [Actions](#) [Create target group](#)

Filter target groups

< 1 >

☐	Name	ARN	Port	Protocol	Target type	Load balancer
☐	AsGroupTG	arn:aws:elasticloadbalancing...	80	HTTP	Instance	AsGroupLB

Step 4 : Verify Deployment

1. Navigate to EC2 → Load Balancers

Load balancers (1) [What's new?](#) [Refresh](#) [Actions](#) [Create load balancer](#)

Filter load balancers

< 1 >

☐	Name	State	Type	Scheme	IP address type	VPC ID
☐	AsGroupLB	Active	application	Internal	IPv4	vpc-0b9967b1715ffa936

2. Copy the DNS name

Load balancers (1/1) [What's new?](#) [Refresh](#) [Actions](#) [Create load balancer](#)

Filter load balancers

< 1 >

☒	Name	State	Type	Scheme	IP address type	VPC ID
☒	AsGroupLB	Active	application	Internal	IPv4	vpc-0b9967b1715ffa936

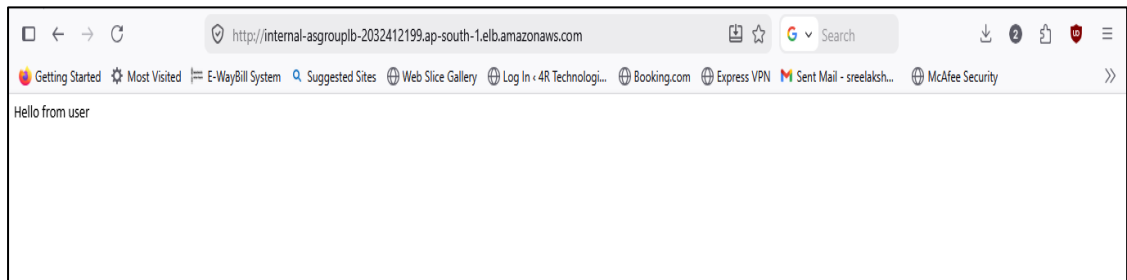
Load balancer: AsGroupLB [Info](#)

Load balancer ARN
[arn:aws:elasticloadbalancing:ap-south-1:965377249696:loadbalancer/ap-south-1/AsGroupLB/9297da827e8dd67f](#)

south-1a (aps-1-az1)

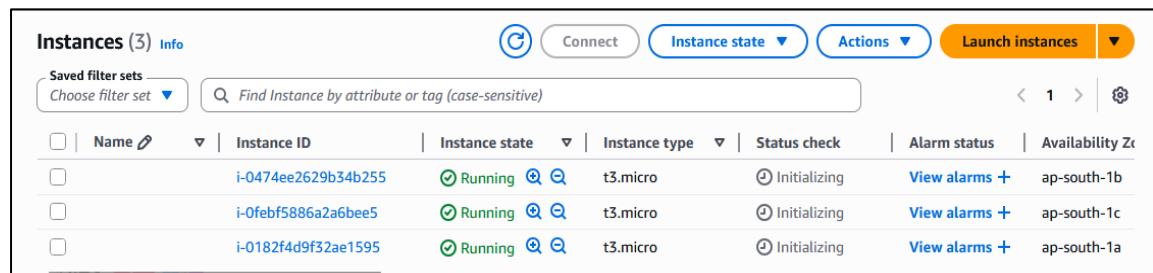
[DNS name Info](#)
[internal-AsGroupLB-2032412199.ap-south-1.elb.amazonaws.com \(A Record\)](#)

3. Open in browser

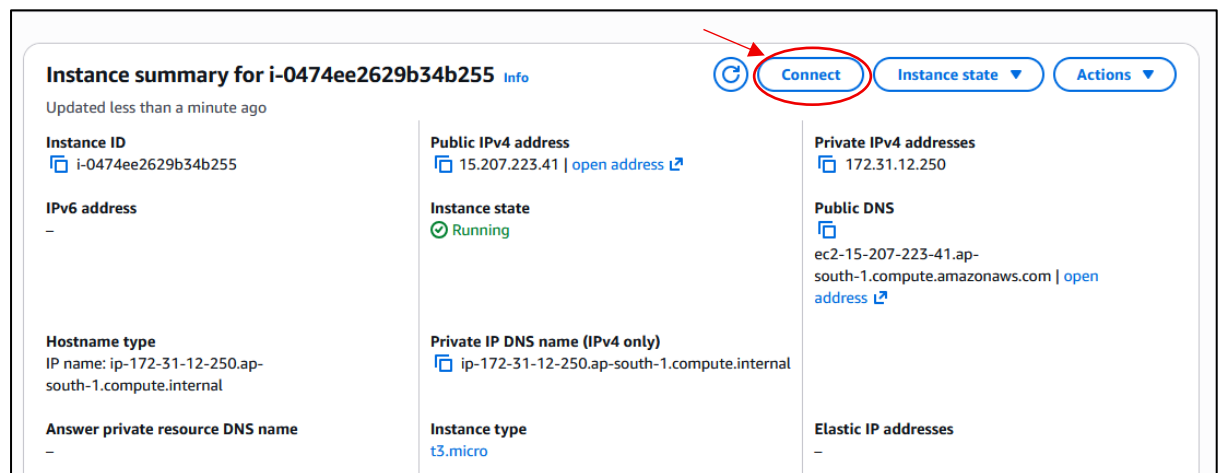


Step 5 : Validate Auto Scaling

1. Navigate to EC2 Instances



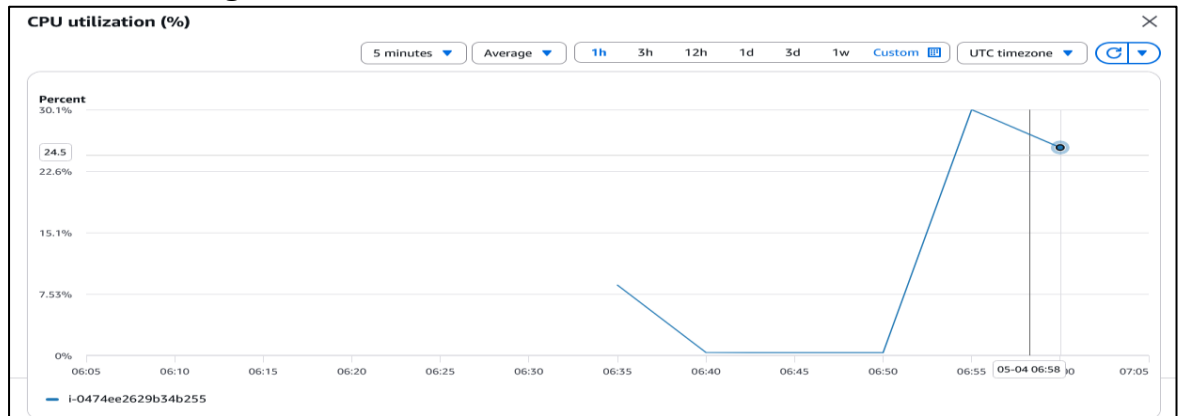
2. Click on any instance and connect



3. Connect to an instance and run:

```
yum install stress -y
stress -c 10 -t 300
```

4. Click **Monitoring** in the instance



Conclusion

This project demonstrated how to deploy a scalable web application using a Launch Template, Auto Scaling Group, and Application Load Balancer on AWS. The system automatically adjusts the number of instances based on CPU utilization and distributes traffic efficiently, ensuring high availability.

Overall, this setup provides a simple yet effective example of how cloud applications can handle varying workloads with minimal manual intervention.